⚡EFFECT OF ELECTRIC CURRENT⚡

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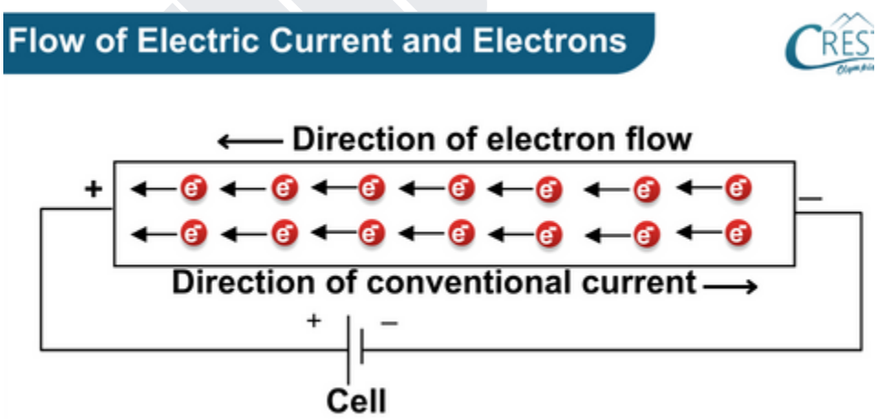
1. Energy Transfer in an Electric Circuit
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1. ⚡ Energy Transfer in an Electric Circuit

- When a charge Q flows from point **A** (higher potential) to point **B** (lower potential) through a resistor, the **cell** supplies energy to the charge.
- The work done on charge Q while moving from A to B is:

$$W = V_{AB} \times Q$$



- This energy is transferred from the **cell** → **charge** → **resistor**, where it is converted into **heat energy**, raising the temperature of the resistor.

🔋 **Electrical Power Formula:**

$$Electrical\ Power = \frac{Electrical\ Energy\ (Work)}{Time}$$

$$P = \frac{V_{AB} \times Q}{t} = V_{AB}.I$$

Unit of Electrical Power:

$$Unit\ of\ POWER\ is\ Watt\ (W)$$

💡 **Key Concept:** If the resistor in the circuit is replaced by a **motor**, electrical energy is converted into **mechanical energy** instead of heat.

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2. 🌡️ Heating Effect of Electric Current

Joule's Law of Heating 🔥

- When current flows through a **resistor**, electrical energy is converted to **heat energy**.
- Heat produced H in a resistor of resistance R, carrying current I for time t:

$$H = P \times t$$

$$H = V_{AB}^2 \times \frac{t}{R}$$

- Applying **Ohm's Law** $(V_{AB} = I \times R)$:

$$H = I^2 \times R \times t$$

💡 Practical Applications of Heating Effect

Device	Heating Element	Material Used	Reason
Electric Heater / Cooker	Coil	**Nichrome** (alloy)	High resistivity, does not melt easily
Electric Bulb	Filament	**Tungsten** wire	Withstands ~3400^oC, emits light
Electric Iron	Plate element	**Nichrome**	Converts electricity to heat efficiently

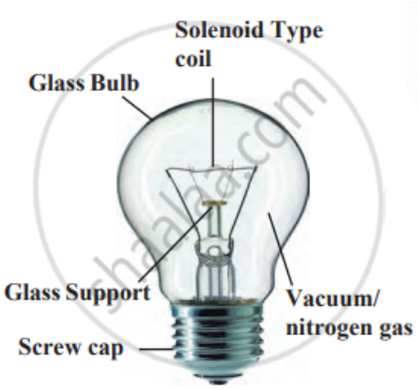
🔌 Units of Electrical Energy

- Practical unit: **kilowatt-hour (kWh)**

$$1kWh = 1000\,W \times 3600\,s = 3.6 \times 10^6\,J$$

JSKY YELE

- **1 Unit of electricity = 1 kWh**





⚠️ Short Circuit & Overloading

Cause	Effect	Safety Device
Short Circuit — live and neutral wires touch due to damaged insulation	Massive current flows → intense heat → fire risk	**Fuse wire** (melts and breaks circuit)
Overloading — too many appliances drawing current beyond transformer capacity	Excessive current → fuse wire melts → supply shuts down	**Fuse wire / MCB**

🔐 **MCB (Miniature Circuit Breaker):** A modern safety switch that automatically opens the circuit when current suddenly increases, replacing traditional fuse wires in homes.





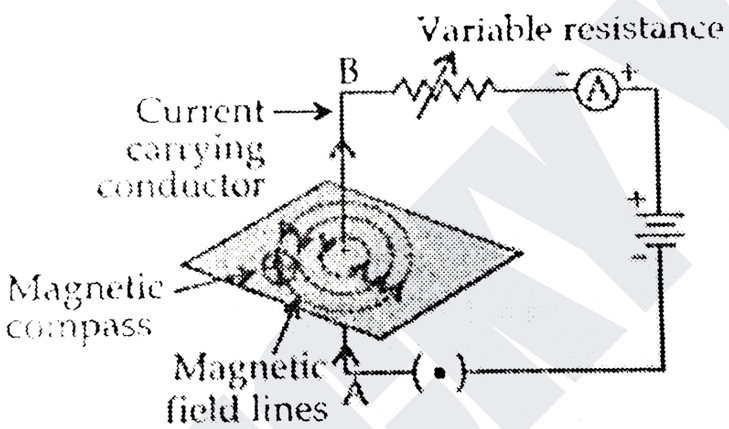
3. 🧲 Magnetic Effect of Electric Current

🧑‍🔬 Introduction — Hans Christian Oersted (1777–1851)

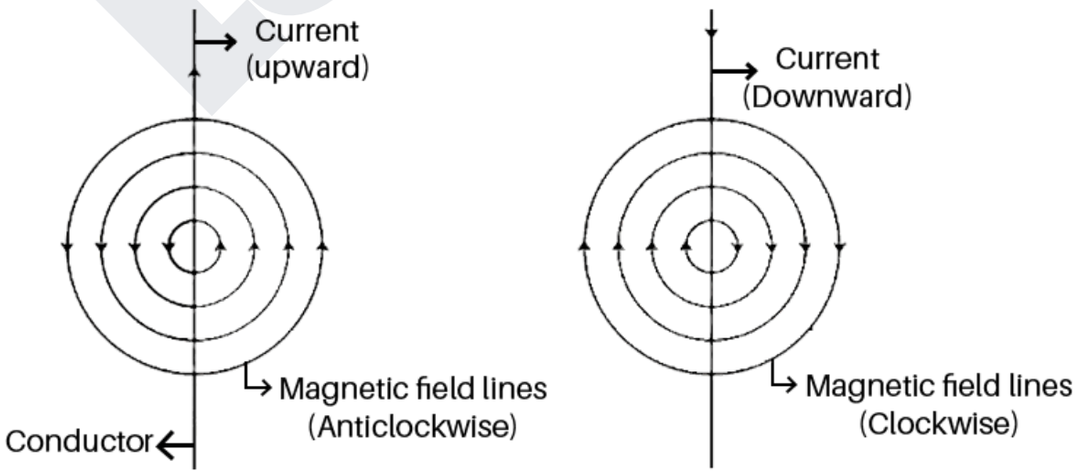
- Hans Christian Oersted, a 19th-century scientist, discovered **electromagnetism** in **1820**.
- He observed that a **magnetic needle deflects** when placed near a metal wire carrying an electric current, linking electricity to magnetism.
- The unit of **intensity of the magnetic field** is named the **Oersted** in his honor.

Key Conclusions from Experiments 🔬

- A **straight current-carrying conductor** produces a **magnetic field around it**.
- The **magnetic lines of force** are **concentric circles** centred on the wire, in a plane perpendicular to the conductor.
- The magnetic field **decreases as distance from the wire increases** (concentric circles become larger and more spread out).
- **Increasing the current** → **increases the intensity** of the magnetic field.
- **Reversing the current direction** → **reverses the direction of the magnetic field**.



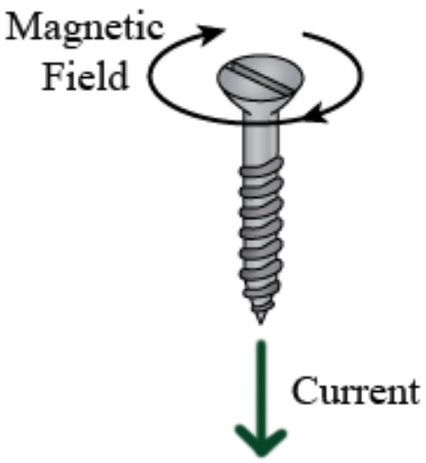
MAGNETIC FIELD PRODUCED BY STRAIGHT CURRENT CARRYING CONDUCTOR ⇓



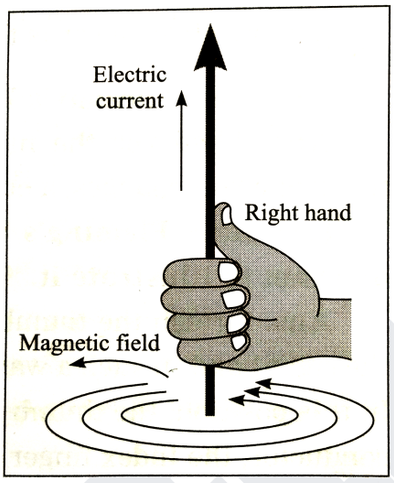
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4. 🤚 Right Hand Thumb Rule

Right Hand Thumb Rule: Imagine holding the current-carrying conductor in your **right hand** such that your **thumb points in the direction of current**. The direction in which your **fingers curl around the conductor** gives the direction of the **magnetic lines of force**.



MAXWELL'S RIGHT HANDED CORK SCREW RULE



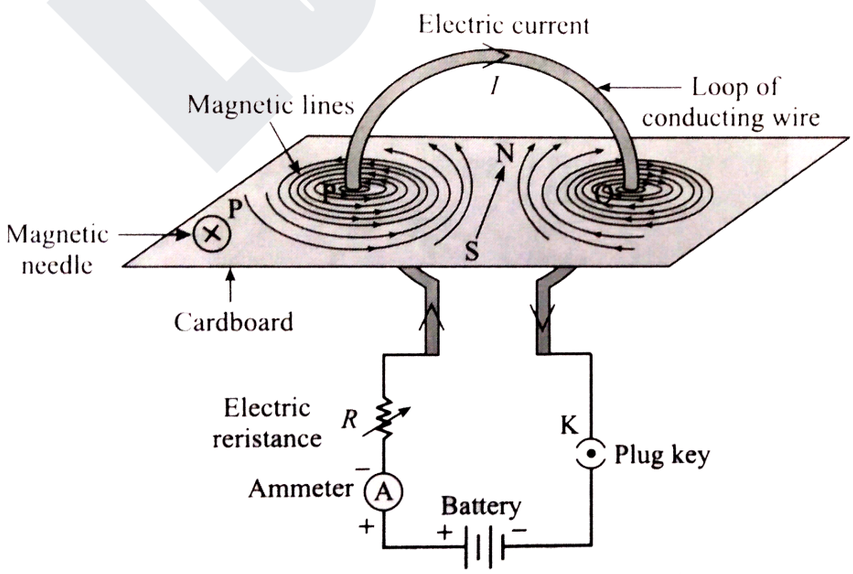
Right hand thumb rule

MAXWELL'S RIGHT HAND RULE

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5. 🔁 Magnetic Field Due to a Current in a Circular Loop

- At every point on the loop, a **circular magnetic field** is produced.
- As we approach the **centre of the loop**, the magnetic circles become so large that their arcs appear as **straight lines**.
- By applying the **Right Hand Thumb Rule** at each point on the loop, all magnetic lines of force at the **centre** are found to be in the **same direction** — they reinforce each other.
- If there are n **turns** in the loop, the magnetic field produced is n **times** that of a single loop.



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6. 🌀 Magnetic Field Due to a Current in a Solenoid

What is a Solenoid?

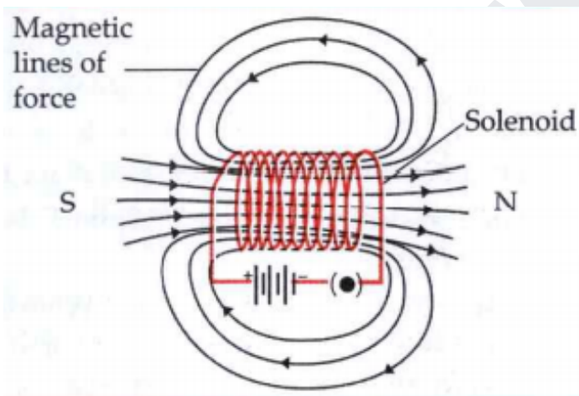
> A **solenoid** is a copper wire (with resistive coating) wound in a **series of loops**, resembling a spring, through which electric current passes.

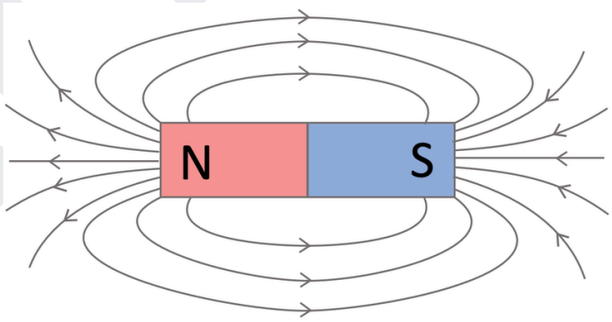
Properties of a Solenoid's Magnetic Field

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Property	Solenoid	Bar Magnet
Shape of field lines	Similar pattern	Similar pattern
Inside the device	**Uniform, parallel** field lines	Uniform field between poles
Poles	One end = **North**, other = **South**	Fixed North and South poles
Field at ends	Diverges outward	Diverges outward

- The magnetic field **inside a solenoid is uniform** — the field lines are parallel to each other throughout.
- A solenoid **behaves like a bar magnet** — one open end acts as the **North pole** and the other as the **South pole**.





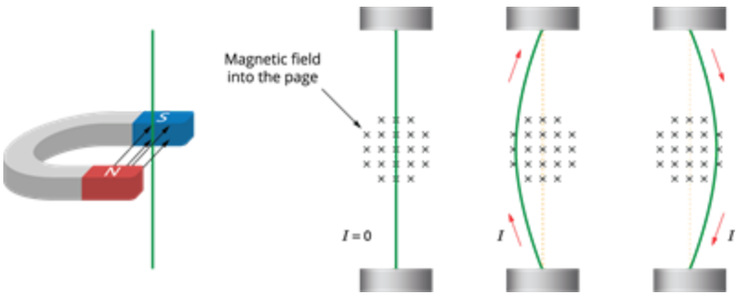
MAGNETIC FIELDS BY CURRENT CARRYING SOLENOID ⇑ *MAGNETIC FIELDS BY BAR MAGNET ⇑*

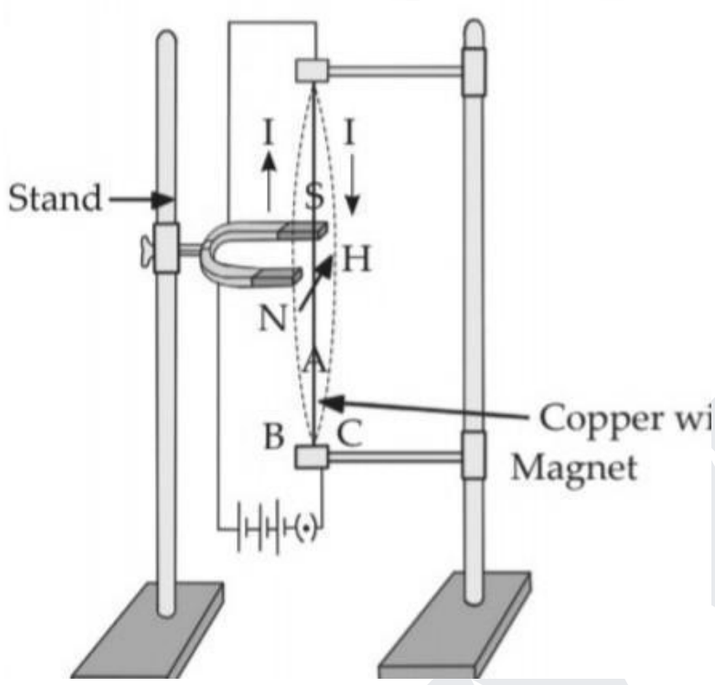
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7. 💥 Force on a Current-Carrying Conductor in a Magnetic Field

Key Conclusions

- When a **current-carrying conductor** is placed in a **magnetic field**, a **force is exerted on it**.
- The direction of this force is **perpendicular to both** the direction of the current and the direction of the magnetic field.
- **Reversing the current direction → reverses the direction of the force**.
- **Reversing the magnetic field direction → reverses the direction of the force**.
- The force is **maximum** when the direction of current is **perpendicular to the magnetic field**.





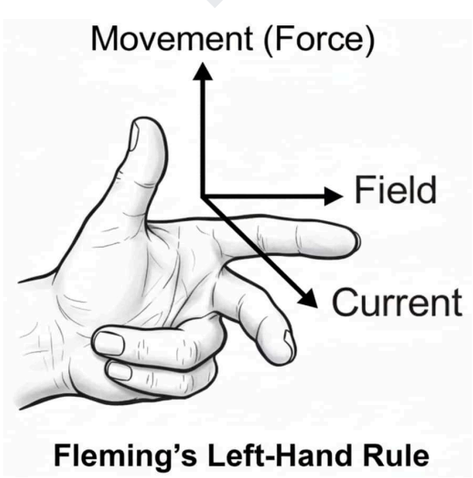
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8. 🤙 Fleming's Left Hand Rule

Fleming's Left Hand Rule: Stretch the **thumb**, **index finger**, and **middle finger** of the left hand so they are **mutually perpendicular**.

- 👆 **Index finger** → direction of the **Magnetic Field (B)**
- 🖕 **Middle finger** → direction of the **Current (I)**
- 👍 **Thumb** → direction of the **Force / Motion of conductor**

🎯 **Mnemonic:** FBI — **F**orce (thumb), **B** field (index), **I** current (middle)

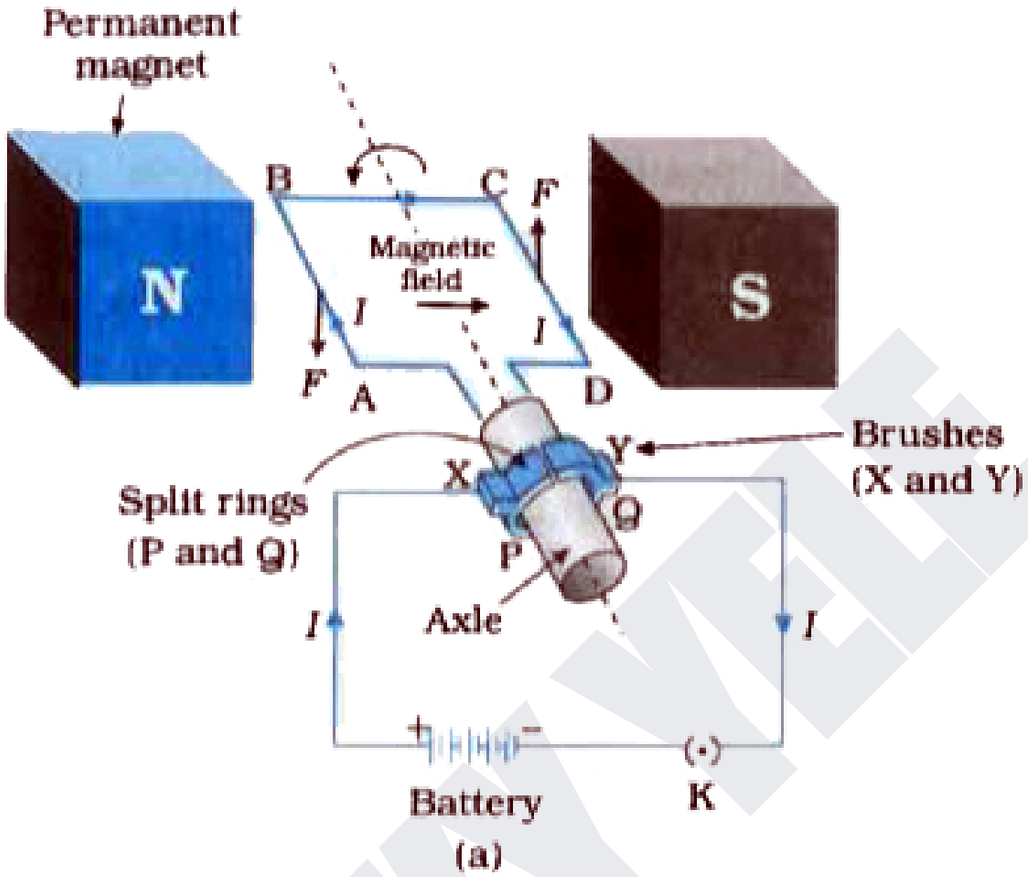


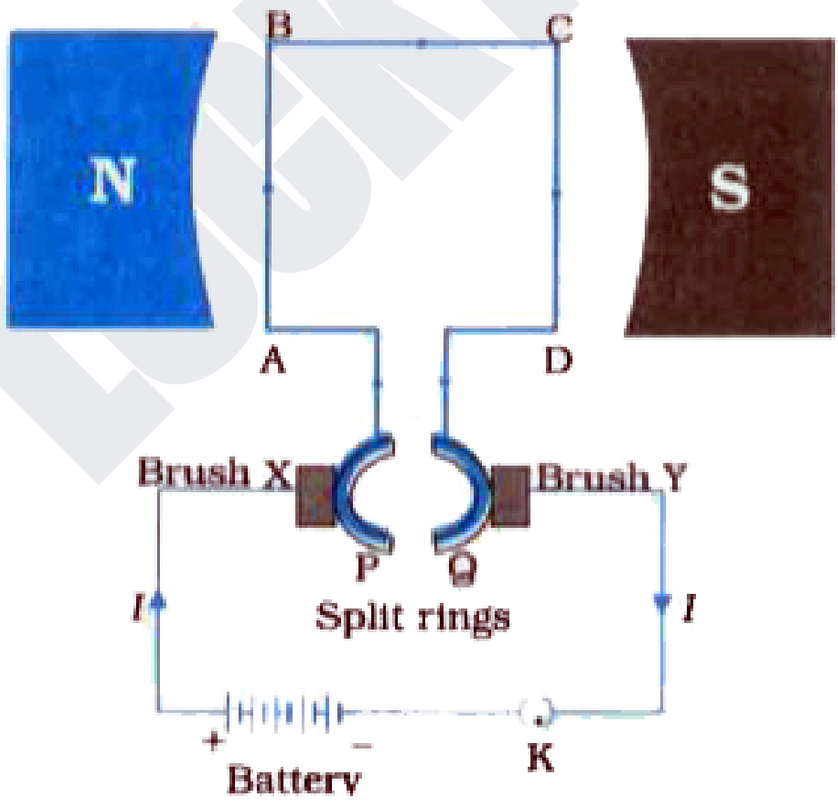
Fleming's Left-Hand Rule

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9. ⚙️ Electric Motor

Definition: An **electric motor** is a device that converts **electrical energy into mechanical energy**.





Common Applications 🏠

Fans, refrigerators, mixers, washing machines, computers, pumps.

Construction

Component	Description
Rectangular coil (ABCD)	Copper wire with resistive coating; placed between magnet poles
Horseshoe Magnet	Provides the external magnetic field
Split Ring (Commutator)	Two halves (X and Y) fixed to the axle; inner surface has resistive coating
Carbon Brushes (E & F)	Stationary; maintain electrical contact with the rotating split ring
Axle	Rotates as the coil turns

Working Principle ⚙️

1. Current enters the coil via carbon brushes → flows through branch **AB** (direction: A → B).
2. By **Fleming's Left Hand Rule**, branch **AB** experiences a **downward force** and branch **CD** experiences an **upward force**.
3. This causes the coil and axle to rotate in an **anticlockwise direction**.
4. After **half a rotation**, the split ring halves swap contact with the brushes → current in the loop **reverses** (now flows D → C → B → A).
5. Forces on AB and CD reverse, but the coil **continues rotating anticlockwise**.
6. The **split ring (commutator)** ensures the current reverses every half rotation, maintaining continuous rotation in one direction.

> 🔑 **Key Principle:** The electric motor works on the principle that **a current-carrying conductor in a magnetic field experiences a mechanical force**.

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10. 🔄 Electromagnetic Induction

👩‍🔬 Introduction — Michael Faraday (1791–1867)

- **Michael Faraday**, a self-taught scientist, started as a teenage bookbinding apprentice.
- Sir Humphrey Davy appointed him as a laboratory assistant at the **Royal Institution, London**.
- He discovered the **laws of electromagnetic induction** in 1831 and the **laws of electrolysis**.
- He humbly declined all honorary degrees offered by various universities.

What is a Galvanometer? 🔬

- A **galvanometer** is a sensitive device that works on the **same principle as an electric motor**.
- A coil is positioned between magnet pole pieces; the pointer is connected to the coil.

- When a **small current** flows through the coil, it rotates **proportionally** to the current.
- The pointer **deflects on both sides** of zero, depending on the **direction of current**.
- **Voltmeter** and **Ammeter** work on the same principle.



Conclusions from Electromagnetic Induction Experiments 🔬

1. When a **conducting wire moves** in a magnetic field, a current is induced in it — this is **Faraday's Electromagnetic Induction**.
2. When a **magnet moves** near a stationary wire, a current is also induced.
3. When current in a **solenoid coil is switched ON or OFF**, a current is induced in a nearby coil.
4. When current in the solenoid is **increased or decreased**, current is induced in the coil.
5. When the **solenoid coil is moved** towards or away from a stationary coil, current is induced.
6. **The faster the displacement** of the solenoid, the **larger the deflection** in the galvanometer (i.e., larger induced current).

⚖️ Faraday's Law of Electromagnetic Induction

Faraday's Law: Whenever the number of **magnetic lines of force (magnetic flux) passing through a coil changes**, a current is **induced** in the coil. The current so produced is called the **induced current**.

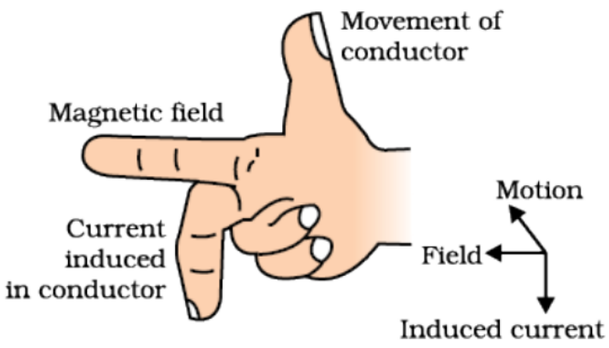
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11. 🤙 Fleming's Right Hand Rule

Fleming's Right Hand Rule: Stretch the **thumb**, **index finger**, and **middle finger** of the **right hand** so they are **mutually perpendicular**.

- 👍 **Thumb** → direction of **Motion of the conductor**
- 👆 **Index finger** → direction of the **Magnetic Field (B)**
- 🖕 **Middle finger** → direction of the **Induced Current**

- The induced current is **maximum** when the direction of motion of the conductor is **perpendicular to the magnetic field**.



⚖️ Comparison: Fleming's Left Hand Rule vs Right Hand Rule

Feature	Fleming's Left Hand Rule	Fleming's Right Hand Rule
Hand Used	Left hand	Right hand
Application	Force on a current-carrying conductor (Motor)	Direction of induced current (Generator)
Thumb	Direction of **Force / Motion**	Direction of **Motion of conductor**
Index finger	Direction of **Magnetic Field**	Direction of **Magnetic Field**
Middle finger	Direction of **Current**	Direction of **Induced Current**
Device	⚙️ Electric **Motor**	🔌 Electric **Generator**

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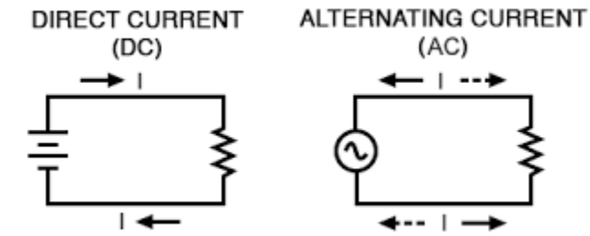
12. Alternating Current (AC) and Direct Current (DC)

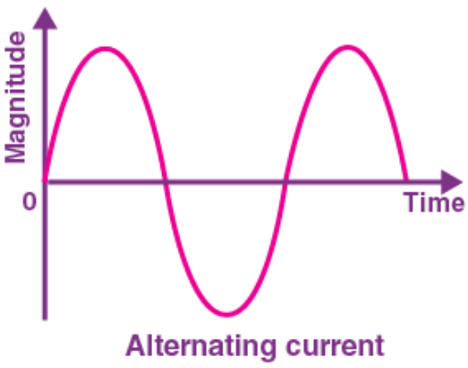
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Feature	Direct Current (DC)	Alternating Current (AC)
Direction	Flows in **one direction only**	**Reverses periodically** (forward & backward)
Nature	Non-oscillatory (can increase, stabilise, or reduce)	**Oscillatory** (sinusoidal)
Graph Shape	Steady / varying but one-directional	**Sinusoidal** wave (symbol: ~)
Pattern	Increases to max → stays / reduces	Increases to max → falls to zero → reverses → max → zero
Source	Cells / Batteries	Power stations, generators
Symbol	——	~

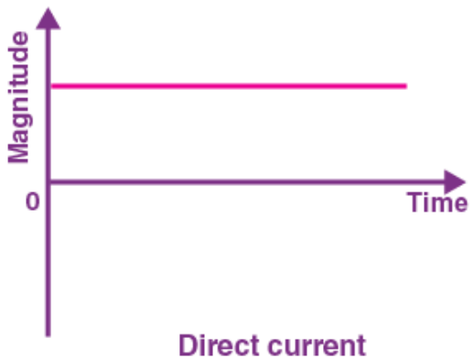
- In India, AC supply has a **frequency of 50 Hz** (one cycle completes in $\frac{1}{50}$ second).

- **AC is preferred for long-distance power transmission** because it results in **minimum power loss**.
- **Home supply** in India is **AC at 220 V**.





Alternating current



Direct current

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13. 🏭 Electric Generator

Definition: An **electric generator** converts **mechanical energy into electrical energy** using the principle of **electromagnetic induction**.

Construction

Component	Description
Coil ABCD	Copper wire coil; rotates between magnet poles
Horseshoe / Bar Magnet	Provides the external magnetic field
Conducting Rings (R_1, R_2)	Fixed to axle (with resistive coating between ring and axle) — used in AC generator
Split Ring (Commutator)	Used in DC generator instead of full rings
Carbon Brushes (B_1, B_2)	Stationary; connected to external circuit (galvanometer)
Axle	Externally rotated by a machine

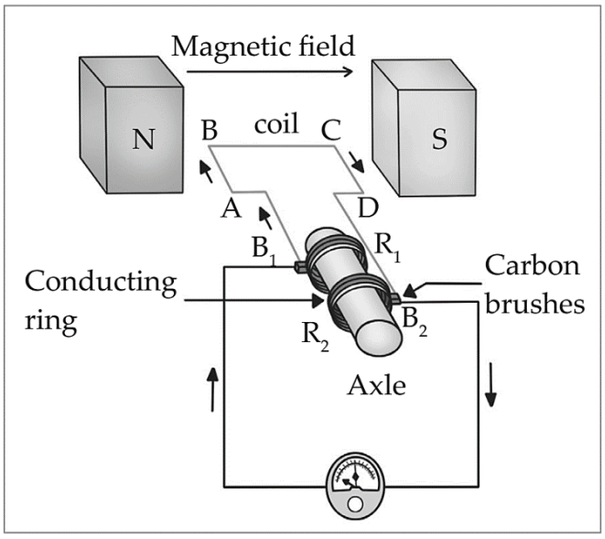
Working — AC Generator ⚡

1. The axle is rotated externally → coil ABCD rotates in the magnetic field.
2. Branch **AB moves up**, branch **CD moves down** → by **Fleming's Right Hand Rule**, current is induced in direction $A \to B \to C \to D$.

3. In the external circuit, current flows from $B_2 \rightarrow B_1$ through the galvanometer.
4. After **half rotation**, AB and CD swap positions → induced current reverses to direction $D \rightarrow C \rightarrow B \rightarrow A$.
5. In external circuit, current now flows $B_1 \rightarrow B_2$ **— opposite to before**.
6. This reversal repeats every half rotation → **Alternating Current (AC) is produced**.
7. Using a **multi-turn coil** instead of a single loop produces a **proportionally larger current**.

Working — DC Generator ⚡

- The **two full conducting rings** are replaced by a **split ring (commutator)**.
- The branch moving **upward** is always in contact with **one brush**, and the branch moving **downward** with the **other brush**.
- This ensures current always flows in **one direction** in the external circuit → **Direct Current (DC)** is produced.



Electric generator

⚖️ AC Generator vs DC Generator

Feature	AC Generator	DC Generator
Output	Alternating Current (AC)	Direct Current (DC)
Ring Type	Full conducting rings (R1, R2)	Split ring (Commutator)
Current Direction	Reverses every half rotation	Always in one direction
Brush Contact	Each brush always touches the same ring	Brush contact switches every half rotation
Application	Power stations, home supply	Charging batteries, industrial uses

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⚡ Class 10 Current Electricity: Ultimate Formula Chart & Study Companion 🔋

This reference document compiles all formulas, derivations, circuit rules, physical constants, and unit conversions into structured, high-yield markdown tables.

📊 1. Fundamental Charge & Current Formulas

Category	Physical Concept	Primary Formula	Mathematical Derivations / Variants	Variable Definitions & Constant Values	SI Unit & Symbol	Application Notes & Pro-Tips
Charge	Quantization of Charge	$Q = ne$	$n = \frac{Q}{e}$	Q = Total electric charge n = Number of electrons ($n \in \mathbb{Z}^+$) e = Elementary charge ($\approx 1.6 \times 10^{-19}$ C)	Coulomb (C)	* Charge cannot exist in fractions of e (e.g., $1.5e$ is impossible). * 1 C of charge contains $\approx 6.25 \times 10^{18}$ electrons.
Current	Electric Current	$I = \frac{Q}{t}$	$I = \frac{ne}{t}$	I = Electric current Q = Charge transferred	Ampere (A)	* 1 A $= 1$ C/s. * Measured using an **Ammeter** connected in **series** (must have near-zero resistance

			$Q = I \cdot t$	t = Time elapsed (in seconds)		
Potential	Electric Potential Difference	$V = \frac{W}{Q}$	$W = V \cdot Q$	V = Potential difference / Voltage	Volt (V)	* $1\ \mathrm{V} = 1\ \mathrm{J/C}$.
			$W = V \cdot I \cdot t$	W = Work done (or energy converted)		* Measured using a **Voltmeter** connected in **parallel** (must have infinitely high resistance to prevent current bypass).
				Q = Test charge		

🌡️ 2. Ohm's Law, Resistance & Physical Dimension Variations

Law / Parameter	Primary Formula	Derived / Variant Formulas	Parameter Definitions	SI Unit (Symbol)	Key Remarks & Conditions
Ohm's Law	$V = IR$	$I = \frac{V}{R}$	V = Voltage across conductor	Ohm (Ω)	* Valid **only** if physical conditions (like **temperature**, mechanical strain, pressure) remain **constant** (Ohmic conductors).
		$R = \frac{V}{I}$	I = Current flowing through		

			Resistance		
Dimensional Factors	$R = \rho \frac{l}{A}$	$R = \rho \frac{l}{\pi r^2}$	R = Resistance	Ohm (Ω)	* $R \propto l$ (longer wire $\rightarrow$ higher resistance).
			ρ = Resistivity of material		* $R \propto \frac{1}{A}$ (thicker wire $\rightarrow$ lower resistance).
		$R = \rho \frac{4l}{\pi d^2}$	l = Length of wire		* Resistivity (ρ) depends **only** on the material nature and temperature, **not** on dimensions.
			A = Cross-sectional area (πr^2 or $\frac{\pi d^2}{4}$)		
			r = Radius, d = Diameter		
Conductance & Conductivity	$G = \frac{1}{R}$	$\sigma = \frac{1}{\rho}$	G = Conductance	Siemens (S or Ω^{-1})	* Conductance is the ease of flow of current.
			σ = Electrical conductivity	$\Omega^{-1}\mathrm{m}^{-1}$	* Conductivity is the reciprocal of resistivity.

🔄 3. Ultimate Stretching, Folding, and Cutting Trick Sheets

mathematical shortcuts:

Mechanical Operation	Length Relation	Area Relation	New Resistance Formula (R′)	Resistivity Relation (ρ′)	Explanation
Stretching by factor n	$l' = n \cdot l$	$A' = \frac{A}{n}$	$R' = n^2 R$	$\rho' = \rho$	Since length increases by n, area must decrease by n to preserve volume. Resistance increases by n^2.
Compressing/Folding n times	$l' = \frac{l}{n}$	$A' = n \cdot A$	$R' = \frac{R}{n^2}$	$\rho' = \rho$	Folding a wire reduces length and increases cross-sectional area. Resistance drops by a factor of n^2.
Radius scaled by factor x	$l' = \frac{l}{x^2}$	$A' = x^2 \cdot A$	$R' = \frac{R}{x^4}$	$\rho' = \rho$	If radius is doubled ($x = 2$), the new resistance becomes $\frac{R}{2^4} = \frac{R}{16}$.
Cutting into n equal parts	$l' = \frac{l}{n}$	$A' = A$	$R' = \frac{R}{n}$	$\rho' = \rho$	Since each piece is cut, volume is not preserved per piece (mass decreases per piece). R' is simply divided by n.

🔌 4. Resistor Combinations (Series vs. Parallel Dualism)

Feature	Series Combination 🔗	Parallel Combination 🔀
Schematic Representation	`---[ R1 ]---[ R2 ]---[ R3 ]---`	`+---[ R1 ]---+` `+---[ R2 ]---+`

```
                                +---[ R3 ]---+
```

Equivalent Resistance (R_{eq})	$R_s = R_1 + R_2 + \cdots + R_n$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
For n Identical Resistors (R)	$R_s = nR$	$R_p = \frac{R}{n}$
Ratio of Extremes ($\frac{R_s}{R_p}$)	$\frac{R_s}{R_p} = n^2$	$\frac{R_p}{R_s} = \frac{1}{n^2}$
Current behavior (I)	Constant through all resistors: $I_{\text{total}} = I_1 = I_2 = I_3$	Divides across parallel branches: $I_{\text{total}} = I_1 + I_2 + I_3$
Voltage behavior (V)	Divides across resistors: $V_{\text{total}} = V_1 + V_2 + V_3$	Constant across all branches: $V_{\text{total}} = V_1 = V_2 = V_3$

	$V_1 = V\left(\frac{R_1}{R_1 + R_2}\right)$	$I_1 = I\left(\frac{R_2}{R_1 + R_2}\right)$
	$V_2 = V\left(\frac{R_2}{R_1 + R_2}\right)$	$I_2 = I\left(\frac{R_1}{R_1 + R_2}\right)$
Overall Value Change	R_s is greater than the largest individual resistance.	R_p is smaller than the smallest individual resistance.
Failure Response	If one component breaks, the entire circuit fails.	If one branch breaks, other branches continue to work normally.

🔥 5. Electric Power, Joule's Heating & Commercial Bill Calculations

Concept	Primary Formula	Equivalent Formulas	Variable Definitions	SI Unit (Symbol)	Practical Application & Pro-Tips

Power	$P = VI$	$P = I^2R$ $P = \frac{V^2}{R}$	Power V = Voltage I = Current R = Resistance		**series** circuits (I is constant). * Use $P = \frac{V^2}{R}$ for **parallel** circuits (V is constant). This is used for domestic rating.
Joule's Heating	$H = I^2Rt$	$H = VIt$ $H = \left(\frac{V^2}{R}\right)t$ $H = P \cdot t$	H = Heat energy generated t = Time of current flow	Joule (J)	* Heating elements (like niches in irons, heaters) use materials with **high resistivity** and **high melting points** (e.g., Nichrome).

Energy	$E = P \cdot t$	$E = V \cdot I \cdot t$	Electrical energy consumed		* **Commercial Unit:** Kilowatt-hour (kWh), also known as "**Unit**".
Commercial Utility Calculation	$\text{Energy (kWh)} = \frac{P(\mathrm{W}) \times t(\mathrm{hr})}{1000}$	$\text{Total Cost} = \text{Energy (kWh)} \times \text{Rate}$	$P(\mathrm{W})$ = Power in Watts $t(\mathrm{hr})$ = Time in hours Rate = Cost per kWh	Currency Units (e.g., ₹ or $)	* **Remember:** You must convert power to **kilowatts** (kW) and time to **hours** (hr) before multiplying to find commercial bills.

6. Unit Conversions & Physical Constants

Physical Quantity	Target Unit	Multiplier / Conversion Factor	Mathematical Representation
Current (I)	Milliampere (mA) to Ampere (A)	Multiply by 10^{-3}	$1\ \mathrm{mA} = 10^{-3}\ \mathrm{A}$
Current (I)	Microampere ($\mu\mathrm{A}$) to Ampere (A)	Multiply by 10^{-6}	$1\ \mu\mathrm{A} = 10^{-6}\ \mathrm{A}$